

PREDICTION IN MULTIVARIATE REGRESSION ANALYSIS

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Summary: The prediction of future observations in the case of multivariate regression has been considered. The sample reuse technique is used to derive a shrunken predictor, and the bootstrap is also considered as a means of obtaining a predictor. Results of simulation studies comparing these predictors with maximum likelihood and other shrunken predictors, Stein and ridge, are also given.

1. INTRODUCTION

Consider a multiple regression problem in which the dependent (Y) and independent variables (Z) (three or more) have a joint normal distribution with unknown mean vector and unknown covariance matrix. Stein (1960) showed that the maximum likelihood estimator of the coefficient vector is inadmissible relative to a quadratic loss function. Baranchik (1964 and 1973) extended Stein's work and proposed minimax estimators for α and β , namely

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$$(1.1) \quad \hat{\beta}_s = \max(0, 1 - c \frac{1-R^2}{R^2}) \hat{\beta}_0$$

and

$$(1.2) \quad \hat{\alpha}_s = \bar{Y} - \hat{\beta}_s \bar{Z}.$$

α and β are the usual parameters in the linear equation $E(Y|Z) = \alpha + \beta Z$, $\hat{\beta}_0$ is the maximum likelihood estimator of β , R is the sample multiple correlation coefficient, c is a non-negative real number bounded by $2(p-2)/(N-p+1)$ where N is the sample size and $p \geq 3$ the number of predictors. $\bar{Y}$ and $\bar{Z}$ are the arithmetic means of the Y and Z sample values respectively. Each of these estimators have lower risks than the maximum likelihood estimators. Zidek (1978) extended Baranchik's results to the multivariate regression problem and obtained minimax estimators for α and β for $q > 1$ dependent variables.

Both the Baranchik-Stein and Zidek-Stein linear prediction functions can be regarded as "flatteners", flattening the usual regression equation by a factor $I-\Lambda$ where $I(q \times q)$ is an identity matrix and the characteristic roots of $\Lambda(q \times q)$ are restricted between 0 and 1. In the case of the Baranchik-Stein prediction function, $I-\Lambda$ is a scalar as defined in equation (1.1).

The purpose of this paper is to find alternative "Stein-type" prediction functions, using the methods of "predictive sample reuse" (Geisser (1975)) and the "bootstrap" (Efron (1978 and 1979)).

An example is considered with three dependent variables and six independent variables.

2. ALTERNATIVE STEIN-TYPE PREDICTION FUNCTIONS

Consider the vector of variables

$$X(p+q \times 1) = \begin{pmatrix} Y \\ Z \end{pmatrix} \begin{matrix} (q \times 1) \\ (p \times 1) \end{matrix} \quad \text{with} \quad E(X) = \mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix} \begin{matrix} (q \times 1) \\ (p \times 1) \end{matrix}$$

$$\text{and } \text{Cov}(XX') = \Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} (p+q \times p+q)$$

For μ and Σ unknown, the object is to predict a future observation of Y for a given Z on a basis of a sample of N past observations of X .

The usual linear regression equation is given by:

$$(2.1) \quad \hat{Y} = \bar{Y} + \hat{\beta}_0(Z - \bar{Z})$$

where $\hat{\beta}_0$ is the least-squares estimator of $\beta = \Sigma_{12} \Sigma_{22}^{-1}$,

i.e. $\hat{\beta}_0(q \times p) = A_{12} A_{22}^{-1}$,

$$A = \sum_{i=1}^N (X_i - \bar{X})(X_i - \bar{X})' = \begin{pmatrix} \sum_{i=1}^N (Y_i - \bar{Y})(Y_i - \bar{Y})' & \sum_{i=1}^N (Y_i - \bar{Y})(Z_i - \bar{Z})' \\ \sum_{i=1}^N (Z_i - \bar{Z})(Y_i - \bar{Y})' & \sum_{i=1}^N (Z_i - \bar{Z})(Z_i - \bar{Z})' \end{pmatrix}$$

$$= \begin{pmatrix} \sum_{i=1}^N (Y_i - \bar{Y})(Y_i - \bar{Y})' & \sum_{i=1}^N U_i \\ \sum_{i=1}^N U_i' & \sum_{i=1}^N V_i \end{pmatrix} \quad \text{say.}$$

$X_i(p+q \times 1)$ denotes the i -th observation of X_i , $Y_i(q \times 1)$ the i -th observation of Y and $Z_i(p \times 1)$ the i -th observation of Z ; $i=1, \dots, N$

$$\bar{X} = \frac{1}{N} \sum_{i=1}^N X_i; \quad \bar{Y} = \frac{1}{N} \sum_{i=1}^N Y_i \quad \text{and} \quad \bar{Z} = \frac{1}{N} \sum_{i=1}^N Z_i.$$

We now specify a linear prediction function of the form:

$$(2.2) \quad \hat{Y} = \bar{Y} + (I - \Lambda) \hat{\beta}_0 (Z - \bar{Z})$$

where Λ is a $q \times q$ matrix of unknown values. Λ is not a parameter of some prior distribution. It is merely a convenient way of forming a prediction function. This function can be regarded as a "Stein-type" prediction function, flattening the usual regression by a factor $I - \Lambda$ where the characteristic roots of Λ are restricted between 0 and 1.

For a sample of N observations, the method of predictive sample reuse (PSR) considers the discrepancy between the i -th observation of Y , $Y_i (q \times 1)$ and the prediction function (2.2) when the remaining $N-1$ observations are used to predict Y_i for given Z_i ; i.e.

$$(2.3) \quad \bar{Y}_{(i)} + (I - \Lambda) \hat{\beta}_{(i)} (Z_i - \bar{Z}_{(i)}) - Y_i .$$

Here $\bar{Y}_{(i)}$, $\bar{Z}_{(i)}$ and $\hat{\beta}_{(i)}$ denote the estimators when the i -th observation is omitted, that is

$$\bar{Y}_{(i)} = \frac{1}{N-1} (N \bar{Y} - Y_i), \quad \bar{Z}_{(i)} = \frac{1}{N-1} (N \bar{Z} - Z_i), \quad \hat{\beta}_{(i)} = A_{12(i)} A_{22(i)}^{-1}$$

where

$$A_{22(i)} = \sum_{\substack{j=1 \\ j \neq i}}^{N-1} (Z_j - \bar{Z}_{(i)}) (Z_j - \bar{Z}_{(i)})' = A_{22} - \frac{N}{N-1} V_i .$$

Also

$$A_{12(i)} = A_{12} - \frac{N}{N-1} U_i .$$

Equation (2.3) can be written as

$$\frac{N}{N-1} \{ (I - \Lambda) \hat{\beta}_{(i)} (Z_i - \bar{Z}) - (Y_i - \bar{Y}) \} .$$

A squared deviation discrepancy where all the Y_i 's are omitted one at a time is given by

$$(2.4) \quad S_N^2(\Lambda) = \frac{N}{(N-1)^2} \sum_{i=1}^N [(I - \Lambda)\hat{\beta}_{(i)}(Z_i - \bar{Z}) - (Y_i - \bar{Y})]' \\ [(I - \Lambda)\hat{\beta}_{(i)}(Z_i - \bar{Z}) - (Y_i - \bar{Y})].$$

The object is to find a good predictive value for Λ , in this case the value that will minimize $S_N^2(\Lambda)$.

Minimization of (2.4) with respect to Λ yields

$$(2.5) \quad \frac{\partial S_N^2(\Lambda)}{\partial \Lambda} = -\frac{2N}{(N-1)^2} \sum_{i=1}^N \{ (I - \Lambda)\hat{\beta}_{(i)}(Z_i - \bar{Z})(Z_i - \bar{Z})'\hat{\beta}_{(i)}' - (Y_i - \bar{Y})(Z_i - \bar{Z})'\hat{\beta}_{(i)}' \}$$

If we put (2.5) equal to 0 (q x q), then it follows that

$$(I - \tilde{\Lambda}) \sum_{i=1}^N \hat{\beta}_{(i)} V_i \hat{\beta}_{(i)}' = \sum_{i=1}^N U_i \hat{\beta}_{(i)}$$

and

$$I - \tilde{\Lambda} = BC^{-1}$$

where

$$B = \sum_{i=1}^N U_i \hat{\beta}_{(i)} \quad \text{and} \quad C = \sum_{i=1}^N \hat{\beta}_{(i)} V_i \hat{\beta}_{(i)}'$$

$\tilde{\Lambda}$ is therefore the PSR estimator of Λ . A solution that restricts the roots of $\tilde{\Lambda}$ to be between 0 and 1 is of interest, when possible.

Remark: This estimator $\tilde{\Lambda}$ will be unaffected by the insertion of $\Sigma_{11.2}^{-1}$ in the discrepancy function (2.4).

The prediction function for a future observation Y for a given Z then becomes

$$(2.6) \quad \hat{Y} = \bar{Y} + \hat{\beta}_p(Z - \bar{Z})$$

where

$$(2.7) \quad \hat{\beta}_p = (I - \tilde{\Lambda})\hat{\beta}_0 = BC^{-1}\hat{\beta}_0$$

and the characteristic roots of $\tilde{\Lambda} = I - BC^{-1}$ may be restricted between 0 and 1. For k future observations the squared prediction error is given by

$$(2.8) \quad S_N^2(\tilde{\Lambda}) = \frac{1}{k} \sum_{j=1}^k \{ \hat{\beta}_p(Z_j - \bar{Z}) - (Y_j - \bar{Y}) \}' \{ \hat{\beta}_p(Z_j - \bar{Z}) - (Y_j - \bar{Y}) \}.$$

Geisser (1975) considered flattening simple regression. The work done so far in this paragraph is therefore an extension of Geisser's work to multiple and multivariate regression.

The bootstrap method, discussed by Efron (1978) and (1979), can be applied to regression analysis as follows:

In this case Λ has to be estimated in such a way that the quadratic difference $\tilde{S}_N^2(\Lambda)$ between $\hat{Y} = \bar{Y} + (I - \Lambda)\hat{\beta}_0(Z - \bar{Z})$ and $E(Y|Z) = \mu_1 + \beta(Z - \mu_2)$ is a minimum. The advantage of the bootstrap is that we can have unknown parameters in the discrepancy function and the Y_i in PSR can therefore be replaced by $E(Y|Z)$.

Minimization of

$$\begin{aligned} \tilde{S}_N^2(\Lambda) = \frac{1}{N} \sum_{i=1}^N \{ \bar{Y} + (I - \Lambda)\hat{\beta}_0(Z_i - \bar{Z}) - (\mu_1 + \beta(Z_i - \mu_2)) \}' \\ \{ \bar{Y} + (I - \Lambda)\hat{\beta}_0(Z_i - \bar{Z}) - (\mu_1 + \beta(Z_i - \mu_2)) \}' \end{aligned}$$

with respect to $\Lambda(q \times q)$ yields

$$(2.9) \quad I - \Lambda^C = \beta A_{22} \hat{\beta}_0' (\hat{\beta}_0 A_{22} \hat{\beta}_0')^{-1}$$

(2.9) however contains the unknown parameter β . By using the bootstrap Λ^C can be estimated as follows:

From our first sample of N observations $X_1, X_2, \dots, X_N$, now called the bootstrap population, draw with replacement a second sample of N observations $X_1^*, X_2^*, \dots, X_N^*$, which is called the bootstrap sample.

Define

$$E(Y^*|Z^*) = \bar{Y} + \hat{\beta}_0(Z^* - \bar{Z}) \quad \text{and} \quad \hat{Y}^* = \bar{Y}^* + \hat{\beta}_0^*(Z^* - \bar{Z}^*)$$

$\bar{Y}$, $\hat{\beta}_0$ and $\bar{Z}$ are now used as the parameters of the bootstrap population.

Minimization of

$$\begin{aligned} \tilde{S}_N^2(\Lambda^*) = & \sum_{i=1}^N \{ \bar{Y}^* + (I - \Lambda^*)\hat{\beta}_0^*(Z_i^* - \bar{Z}^*) - (\bar{Y} + \hat{\beta}_0(Z_i^* - \bar{Z})) \}^2 \\ & \{ \bar{Y}^* + (I - \Lambda^*)\hat{\beta}_0^*(Z_i^* - \bar{Z}^*) - (\bar{Y} + \hat{\beta}_0(Z_i^* - \bar{Z})) \} \end{aligned}$$

with respect to $\Lambda^*(q \times q)$ yields

$$(2.10) \quad I - \Lambda^* = \hat{\beta}_0 A_{22}^* \hat{\beta}_0^{*'} (\hat{\beta}_0^* A_{22}^* \hat{\beta}_0^{*'})^{-1}.$$

This may be called the bootstrap estimator of (2.9).

$\bar{Y}^*$, $\bar{Z}^*$, $\hat{\beta}_0^*$ and A_{22}^* are obtained in the same way from the bootstrap sample as $\bar{Y}$, $\bar{Z}$, $\hat{\beta}_0$ and A_{22} were from the original sample.

The bootstrap procedure can be repeated a large number of times, say L .

$$(2.11) \quad \Lambda^* = \frac{1}{L} \sum_{k=1}^L \Lambda_k^* \quad \text{can then be used as an estimate of } \Lambda.$$

3. SIMULATION STUDY

A simulation study was undertaken to compare the PSR, bootstrap and ridge predictors or estimators with the maximum likelihood. The approach was Bayesian, and in the first case, case A, a $N(\underline{0}; I_9)$ parent population was used to generate populations for sampling. For each simulation, one random observation was drawn from the parent population to constitute the mean vector $\mu(9 \times 1)$, and another random sample of 15 determined the covariance matrix Σ , i.e.

$$\Sigma(9 \times 9) = \frac{1}{15} WW', \text{ where } W(9 \times 15) \sim N(0, I_9 \otimes I_{15}).$$

Three dependent and 6 independent variables were considered so that for each simulation;

$$X = \begin{pmatrix} Y \\ Z \end{pmatrix} \begin{matrix} (3 \times 1) \\ (6 \times 1) \end{matrix} \sim N(\mu, \Sigma).$$

From this population a sample of 16 observations was drawn and the first 15 used to calculate the different prediction functions which were then used to predict the last observation, Y , for given Z . The quadratic loss was then calculated.

In case B, the covariance matrix of the parent population was changed to Φ , where

$$\Phi = \begin{pmatrix} 16 & -8 & 0 & 16 & 0 & -8 & 0 & -16 & 8 \\ & 8 & 2 & -14 & -4 & 8 & 2 & 12 & -8 \\ & & 11 & -8 & -2 & 7 & 16 & 22 & -7 \\ & & & 62 & 24 & -21 & -16 & -50 & 39 \\ & & & & 113 & -7 & -38 & -56 & 46 \\ & & & & & 59 & -2 & 34 & -20 \\ & & & & & & 25 & 3 & -25 \\ & & & & & & & 269 & -24 \\ & & & & & & & & 120 \end{pmatrix}$$

In this case there was a much higher degree of multicollinearity between independent variables and a stronger correlation between Y and Z than in case A where the regression coefficients would, in general, be close to zero. The large variances in Φ also ensured a wide range of μ and Σ values in case B.

The simulations were repeated 5000 times in both cases and the following regression coefficient estimators calculated:

(i) The maximum-likelihood estimator,

$$(3.1) \quad \hat{\beta}_0(3 \times 6) = A_{12} A_{22}^{-1}, \text{ where } A \text{ is given below (2.1).}$$

(ii) The Zidek-Stein estimator (Zidek (1978)), given by

$$(3.2) \quad \begin{aligned} \hat{\beta}_S &= \{I_q - \frac{1}{n-p-q} A_{11.2}^{-1} (A_{12} A_{22}^{-1} A_{21})^{-1}\} \hat{\beta}_0 \\ &= \{I_3 - \frac{1}{6} A_{11.2}^{-1} (A_{12} A_{22}^{-1} A_{21})^{-1}\} \hat{\beta}_0 \text{ for this} \\ &\text{example with } A_{11.2} = A_{11} - A_{12} A_{22}^{-1} A_{21} . \end{aligned}$$

(iii) The PSR estimator derived in section 2 and given in (2.7), but without restricting the roots to the interval $[0,1]$. That is,

$$(3.3) \quad \hat{\beta}_p = BC^{-1} \hat{\beta}_0 .$$

(iv) The modified PSR estimator with the roots of $I-BC^{-1}$ adapted so as to lie in the interval $[0,1]$. 7,56% of the simulations in case A and 9,28% in case B, gave complex roots and in these cases no modifications were applied. This estimator is denoted by $\hat{\beta}_p^m$.

- (v) In trying to solve the problem of complex roots for the previous estimator, a symmetrical version of (3.3) was examined, namely

$$(3.4) \quad \hat{\beta}_{PS}^m = \frac{1}{2}(BC^{-1} + C^{-1}B')\hat{\beta}_0$$

with the roots of $I - \frac{1}{2}(BC^{-1} + C^{-1}B')$ restricted to the interval $[0,1]$.

- (vi) The estimator

$$(3.5) \quad \hat{\beta}_C = (I - \Lambda^C)\hat{\beta}_0, \text{ with } I - \Lambda^C \text{ given in (2.9), was}$$

also considered. This value of $I - \Lambda$ minimizes the discrepancy function $\tilde{S}_N^2(\Lambda)$ for a given sample when $\beta = \Sigma_{12}\Sigma_{22}^{-1}$ is known. It is the value that the bootstrap estimator (2.10) tries to estimate. $\hat{\beta}_C$ is included for comparison purposes only.

- (vii) The bootstrap estimator, namely

$$(3.6) \quad \hat{\beta}_B = (I - \Lambda^*)\hat{\beta}_0, \text{ with } I - \Lambda^* \text{ given in (2.10). A}$$

bootstrap sample of 15 was drawn from each sample and $I - \Lambda^*$ calculated. To stabilize the estimator this was repeated 5 times and the mean taken according to (2.11).

- (viii) The ridge regressor included in the simulation is the so-called modified Hoerl-Kennard-Baldwin rule from Brown and Zidek (1979), namely

$$(3.7) \quad \hat{\beta}_R = \hat{\alpha}^*P$$

where the i -th row of $\hat{\alpha}^*$ is given by

$$(3.8) \quad \hat{\alpha}_i^* (1 \times q) = \hat{\alpha}_i [I_q - \frac{p-q-1}{\lambda_i} \{ \frac{p-q-1}{\lambda_i} I_q + (n-q+1) A_{11.2}^{-1} \sum_{j=1}^p \hat{\alpha}_j^* \hat{\alpha}_j \}^{-1}] i=1, \dots, p$$

and $\hat{\alpha}_i$ is the i -th row of

$$(3.9) \quad \hat{\alpha} = P \hat{\beta}_0'$$

where

$$P A_{22} P' = \text{Diag}(\lambda_1, \dots, \lambda_p) \quad \text{and} \quad P' P = I_p.$$

The prediction functions are of the form (2.1), namely

$$\hat{Y} = \bar{Y} + \hat{\beta}(Z_K - \bar{Z})$$

where $Z_K(6 \times 1)$ represents the independent part of the 16th observation. $Y_K(3 \times 1)$ was predicted by (2.1) using each of the eight regression coefficients given above. The following two quadratic loss functions were considered:

$$(3.10) \quad S_1^2 = (\hat{Y} - Y_K)' (\hat{Y} - Y_K).$$

and

$$(3.11) \quad S_2^2 = (\hat{Y} - Y_K)' \Sigma_{11.2}^{-1} (\hat{Y} - Y_K).$$

$\bar{S}_1^2$ and $\bar{S}_2^2$ are the averages obtained from 5000 simulations.

The following results were obtained.

TABLE 3.1

Case A

Predictor	$\hat{\beta}_0$	$\hat{\beta}_S$	$\hat{\beta}_P$	$\hat{\beta}_P^m$	$\hat{\beta}_{PS}^m$	$\hat{\beta}_C$	$\hat{\beta}_B$	$\hat{\beta}_R$
$\bar{S}_1^2$	3,5073	3,2812	3,0621	2,9118	2,9025	2,3949	3,2228	3,2184
$\bar{S}_2^2$	5,9406	5,6362	5,4255	5,1939	5,4872	4,2270	5,5858	5,5184

Case B

Predictor	$\hat{\beta}_0$	$\hat{\beta}_S$	$\hat{\beta}_P$	$\hat{\beta}_P^m$	$\hat{\beta}_{PS}^m$	$\hat{\beta}_C$	$\hat{\beta}_B$	$\hat{\beta}_R$
$\bar{S}_2^2$	23,2936	22,2652	22,2336	21,6579	22,6718	19,9244	22,6415	21,9727
$\bar{S}_2^2$	5,8682	5,6749	5,8651	5,7239	8,0868	5,3345	5,8128	5,6230

Remarks:

- (i) The "correct" prediction function gave the smallest mean quadratic loss. This was expected since the true value of β was used in $\hat{\beta}_C$.
- (ii) Except for $\bar{S}_2^2$ of $\hat{\beta}_{PS}^m$ in case B, the six shrunken predictors had smaller quadratic loss than the maximum likelihood predictor. The magnitude of the improvement will however be determined by the design of the experiment. In case A the parent population was $N(0,1_g)$ with the consequence that the regression coefficients of the different simulations will in general be close to zero and thus favour a shrunken predictor. Thus the size of the improvement is throughout larger in case A than in case B.

(iii) In case A the PSR predictors using $\hat{\beta}_p, \hat{\beta}_p^m$ and $\hat{\beta}_{ps}^m$ performed the best. The modified PSR predictor $\hat{\beta}_p^m$ improved on $\hat{\beta}_p$ in both cases. Although the "symmetrical" version $\hat{\beta}_{ps}^m$ did quite well in case A it showed a dramatic increase in quadratic loss for $\bar{S}_2^2$ in case B. This casts serious doubt on the usefulness of $\hat{\beta}_{ps}^m$.

(iv) In case A the ridge as well as the bootstrap predictor improved on the Stein predictor. In fact the Stein predictor seems to be the worst of the shrunken predictors in this case.

(v) The modified PSR predictor $\hat{\beta}_p^m$ also did well in case B, being the best of all the shrunken predictors for $\bar{S}_1^2$. In this case, because of the higher degree of multicollinearity, the ridge predictor came into its own being the best for $\bar{S}_2^2$ and second best for $\bar{S}_1^2$.

Stein also improved in case B, beating the bootstrap for both loss functions and PSR for $\bar{S}_2^2$.

(vi) In another simulation study which was conducted by Groenewald, Van der Merwe and De Waal (1980) a real multiple regression problem was used. The data were obtained from Anderson and Bancroft (1952). The independent variables were

$Z_1 \equiv$ Percentage of nitrogen	$Z_4 \equiv$ Percentage of phosphorus
$Z_2 \equiv$ Percentage of chlorine	$Z_5 \equiv$ Percentage of calcium
$Z_3 \equiv$ Percentage of potassium	$Z_6 \equiv$ Percentage of magnesium

The dependent variable $Y \equiv$ Rate of cigarette-burn in inches per 1000 seconds. In this case a sample of 15 was drawn from a normal population with mean and covariance

matrix equal to that of the sample of real data.

The following results were obtained for 1000 simulations.

TABLE 3.2

Predictor	$\hat{\beta}_0$	$\hat{\beta}_S$	$\hat{\beta}_P^m$	$\hat{\beta}_C$	$\hat{\beta}_B$
$\bar{S}_1^2$	0,01723	0,01632	0,01613	0,01482	0,01636

As in the previous case, the "correct" prediction gave the smallest mean quadratic loss. Among the rest, the PSR predictor gave the best performance. It was also the predictor that was most frequently closest to the value being predicted.

- (vii) The conclusion that can be drawn from the simulation studies is that the method of predictive sample reuse performs just as well or even better than well-known shrunken predictors such as Stein or ridge in the case of multivariate regression analysis.

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REFERENCES

- ANDERSON, R.L. and BANCROFT, T.A. (1952). *Statistical Theory in Research* (page 205). McGraw-Hill, Book Company, Inc.

- BARANCHIK, A.J. (1964). Multiple Regression and Estimation of the Mean of a Multivariate Normal Distribution. *Technical Report, No. 51, Dept. of Statistics, Stanford University, Stanford, California.*
- BARANCHIK, A.J. (1973). Inadmissibility of Maximum Likelihood Estimators in Some Multiple Regression Problems with Three or More Independent Variables. *Ann. of Statist.* 2, 312-321.
- BROWN, P.J. and ZIDEK, J.V. (1979). Multivariate Ridge Regression with unknown Covariance Matrix. *Technical Report No. 79-11, The Institute of Applied Mathematics and Statistics, University of British Columbia, Vancouver.*
- EFRON, B. (1978). Computers and the Theory of Statistics: Thinking the Unthinkable. *Technical Report No. 39, Dept. of Statistics, Stanford University, Stanford, California.*
- EFRON, B. (1979). Bootstrap Methods: Another Look at the Jackknife. *Ann. of Statist.* 7, 1-26.
- GEISSER, S. (1975). The Predictive Sample Reuse Method with Applications. *J. Amer. Statist. Assoc.*, 70, 320-328.
- GROENEWALD, P.C.N., VAN DER MERWE, A.J. and DE WAAL, D.J. (1980). Prediction and Modelselection in Multivariate Regression Analysis. *Technical Report No. 61, Dept. of Math. Statistics, University of the Orange Free State.*
- STEIN, C. (1960). Multiple Regression, in J. Olkin (ed.) *Contributions to Probability and Statistics. Essays in Honor of Harold Hotelling*, Stanford University Press, Stanford, California, 424-443.
- ZIDEK, J.V. (1978). Deriving Unbiased Risk Estimators of Multinormal Mean and Regression Coefficient Estimators using Zonal Polynomials. *Ann. of Statist.* 6, 769-782.